Ex.No: 19 Date:

Study of Topology - Benchmark LAN Trainer Kit

AIM:

To study the various network topologies - bus, star, ring, mesh, tree and hybrid- and to physically construct and observe them using the Benchmark LAN Trainer Kit.

SOFTWARE REQUIRED:

• Benchmark LAN Trainer Kit with its node modules, central hub/switch module and patch cords.
• Set of connecting cables (RJ-45 patch cords or the kit's supplied connectors). • Kit power supply and the accompanying observation/indicator panel.

THEORY:

Network topology is the physical or logical arrangement in which nodes (computers, terminals or the kit's node modules) are interconnected. The topology determines how data flows, how much cabling is needed, how the network behaves when a link or node fails, and how easy it is to expand. A LAN trainer kit provides node modules and a central connecting module so that each topology can be wired up by hand and its behaviour — signal path, fault effect, expandability — directly observed on the kit's indicator panel.

Addressing plan

Topology Arrangement Advantages Disadvantages

Bus All nodes share one common backbone cable with terminators at both ends.

Simple; least cabling; cheap.

One cable fault stops the whole network; hard to troubleshoot.

Star Every node connects to a central hub/switch.

Easy to add/remove nodes; single node fault does not affect others.

Central device failure downs the whole network; more cable.

Ring Nodes form a closed loop; data passes node to node.

Orderly, predictable data flow; no collisions with token passing.

A single break can halt the ring; adding nodes disrupts it.

Mesh Every node links directly to every other node.

Highly reliable; multiple paths; faulttolerant.

Very expensive; huge amount of cabling.

Tree Groups of star networks joined to a bus backbone (hierarchy).

Scalable; good for large structured networks.

Backbone failure affects large sections; complex.

Hybrid Combination of two or more of the above.

Flexible; tailored to needs.

Design and management are complex and costly.

PROCEDURE:

1. Place the LAN Trainer Kit on the bench and connect its power supply. Switch it on and confirm the indicator lamps/LEDs are functioning.

2. Identify the components: the individual node modules, the central hub/switch module, the backbone connector, and the supplied patch cords.

Bus topology

3. Connect all node modules along the single common backbone line provided on the kit, fitting the terminators at both ends of the bus.

4. Send data from one node and observe on the indicator panel that the signal travels along the shared backbone and is seen by all nodes; only the addressed node accepts it.

5. Disconnect the backbone at a mid-point and note that communication beyond the break stops — demonstrating the bus's single point of failure.

Star topology

6. Connect each node module by a separate patch cord to a port on the central hub/switch module.

7. Transmit from one node and observe that data passes through the central module to reach the destination node.

\ 8. Disconnect one node's cable and confirm the other nodes continue to communicate; then power off the central module and note that the whole network stops — showing the star's dependence on the central device.

Ring topology

9. Wire the node modules in a closed loop — the output of each node to the input of the next — until the last node connects back to the first.

10. Send data and observe it circulating from node to node around the ring until it reaches the intended destination.

11. Open the loop at one link and observe that the ring's data flow is interrupted.

Mesh topology

12. Using the available patch cords, connect each node module directly to every other node module.

13. Transmit between two nodes and observe that, because multiple direct paths exist, communication continues even if one link is removed — demonstrating fault tolerance.

14. For each topology, record on the observation sheet: number of cables used, the path taken by the data, and the effect of a single link or node failure.

SAMPLE OBSERVATION TABLE:

Topology

Cables used (for n nodes)

Effect of one link failure

Ease of adding a node

Bus n + backbone Whole segment may fail Easy

Star n Only that node affected Very easy

Ring n Ring flow interrupted Difficult

Mesh n(n−1)/2 No effect (alternate paths) Difficult

RESULT:

Thus, the bus, star, ring and mesh network topologies were studied and physically constructed on the Benchmark LAN Trainer Kit.